B BELLOMASSI

SYSTEM REQUIREMENTS

Functional Requirements Document (Consolidated)

Bellomassi Order-to-Delivery Management Platform

Document ID BM-SYS-003

Document control

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> **Status labels used in this document:** VERIFIED (confirmed in the current source code or existing documents) · PROTOTYPE (works in the React/LocalStorage prototype with mock or simulated behaviour) · PROPOSED (recommended production design, not built) · PENDING CONFIRMATION (needs a client decision) · NOT IMPLEMENTED · NOT VERIFIED.

Purpose

List every functional requirement of the platform with a unique ID, priority, expected result and implementation label.

Scope

All 35 modules; detailed specifications per module are in folder 04.

Definitions

Other terms: see **BM-GLO-001** (Project Glossary).

References

- **BM-SYS-002** (Software Requirements Specification (SRS))
- **BM-SYS-008** (Requirement Traceability Matrix)
- **BM-BRD-001** (Business Requirements Document (BRD))

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1. Requirement summary

Module	FRs	VERIFIED	PROTOTYPE	PROPOSED	PENDING	NOT IMPL.	NOT VERIFIED
Authentication and Login (AUTH)	13	10	1	2	0	0	0
User Management (USR)	5	5	0	0	0	0	0

Module	FRs	VERIFIED	PROTOTYPE	PROPOSED	PENDING	NOT IMPL.	NOT VERIFIED
Role and Permission Management (RBAC)	6	5	0	1	0	0	0
Dashboards and Analytics (DASH)	6	5	0	1	0	0	0
Customer Management (CUS)	5	4	0	0	1	0	0
Product Catalogue (PRD)	4	3	0	1	0	0	0
Hubs, Suppliers and Organisation Master (HUB)	3	3	0	0	0	0	0
Salesman Order Management (ORD)	11	9	0	0	1	1	0
Sales Manager Approval (APR)	8	6	0	0	1	1	0
Credit Validation (CRD)	5	3	0	1	0	1	0
Order Consolidation (Customer Service) (CON)	4	4	0	0	0	0	0
Dispatch Management (DSP)	9	7	0	1	1	0	0
Zone Management (ZON)	2	2	0	0	0	0	0
Driver and Vehicle Assignment (DRA)	4	4	0	0	0	0	0
Driver Application (DRV)	11	9	0	2	0	0	0
Customer Application (CAP)	8	6	0	0	1	1	0
Delivery Tracking (TRK)	3	1	1	1	0	0	0
Delivery OTP (OTP)	5	3	1	1	0	0	0
Proof of Delivery (POD)	3	2	0	1	0	0	0
Delivery Notes (DN)	3	2	0	1	0	0	0
Delivery Exception Review (EXC)	3	3	0	0	0	0	0
Inventory Management (INV)	8	6	0	1	0	1	0
Scrap Management (SCR)	5	5	0	0	0	0	0
Warehouse and QA Dashboards (WQA)	3	2	0	0	0	1	0
Vehicle Management System (VMS)	5	4	0	0	0	1	0
Document Management and OCR (DOC)	5	3	1	1	0	0	0
Invoice Customisation and Invoicing (INVC)	6	4	0	0	1	1	0
Email Communication (EML)	4	1	2	0	0	0	1
OpenWA (WhatsApp) Communication (OWA)	3	0	1	0	1	0	1
Automated Notifications (AUTO)	2	2	0	0	0	0	0
Call History (CALL)	2	2	0	0	0	0	0
Notifications (in-app) (NOTF)	3	2	0	1	0	0	0
Reports (RPT)	3	3	0	0	0	0	0
AI Assistant, Forecasting and Replenishment (AI)	5	3	1	0	0	0	1
System Settings and Audit (SET)	4	3	0	1	0	0	0
Total	**179**	136	8	17	7	8	3

Authentication and Login (AUTH)

Detail: **BM-FRD-AUTH** (Functional Requirements — Authentication and Login)

ID	P	Requirement	Expected result	Label
FR-AUTH-001	P1	The system shall authenticate office users by email or username and password and open the role landing page.	Valid credentials open the landing page of the user's role.	VERIFIED
FR-AUTH-002	P1	The driver app shall accept mobile number, email or username as the identifier.	Driver signs in with +91 98400 11206 / Driver@123 and lands on /driver/dashboard.	VERIFIED

ID	P	Requirement	Expected result	Label
FR-AUTH-003	P1	The customer app shall accept mobile number, email or username as the identifier.	Customer signs in with +91 98410 22310 / Customer@123 and lands on /customer/dashboard.	VERIFIED
FR-AUTH-004	P1	The system shall lock an account for the configured lockout period after the configured number of failed attempts (default 5 attempts, 15 minutes).	Sixth attempt shows “Too many failed attempts. This account is locked for 15 minutes.”	VERIFIED
FR-AUTH-005	P1	The system shall block sign-in for inactive accounts with a clear message.	“This account is inactive. Contact your administrator to reactivate it.”	VERIFIED
FR-AUTH-006	P1	The system shall sign users out after the configured idle period (default 60 minutes).	Session ends after inactivity; user returns to the sign-in page of the same app.	VERIFIED
FR-AUTH-007	P2	“Keep me signed in” shall persist the session across browser restarts; otherwise the session ends with the tab.	Session stored in localStorage when ticked, sessionStorage otherwise.	VERIFIED
FR-AUTH-008	P2	Users shall reset a forgotten password with a 6-digit code valid for 15 minutes.	Code is displayed on screen in the prototype (no email is sent).	PROTOTYPE
FR-AUTH-009	P1	Users shall be forced to change their password after an administrator reset when the setting is enabled.	User is redirected to /account/change-password?required=1 until the password is changed.	VERIFIED
FR-AUTH-010	P1	Sign-out shall return the user to the sign-in page of the app they were using.	Driver → /driver/login, customer → /customer/login, office → /login.	VERIFIED
FR-AUTH-011	P1	Every route shall require sign-in and the route’s permission; unauthorised users see the Unauthorized page.	Opening /admin/users as Salesman shows /unauthorized.	VERIFIED
FR-AUTH-012	P1	Production authentication shall use server-side hashed passwords (Argon2/PBKDF2) and short-lived JWT access tokens with refresh tokens.	No credentials or password hashes are stored in the browser.	PROPOSED
FR-AUTH-013	P2	Privileged roles shall use multi-factor authentication (TOTP) in production.	Admin sign-in requires a second factor.	PROPOSED

User Management (USR)

Detail: **BM-FRD-USR** (Functional Requirements — User Management)

ID	P	Requirement	Expected result	Label
FR-USR-001	P1	Administrators shall create users with name, username, email, phone, role, home hub and (for portal users) the linked customer, driver or supplier.	User appears in the list and can sign in.	VERIFIED
FR-USR-002	P1	Administrators shall activate, deactivate, unlock and reset passwords of users.	Status changes are effective at the next sign-in and recorded in the audit log.	VERIFIED
FR-USR-003	P1	The system shall prevent users from deactivating their own account and only a Super Admin shall change a Super Admin.	“You cannot deactivate your own account.” / “Only a Super Admin can change a Super Admin account.”	VERIFIED
FR-USR-004	P2	Administrators shall assign roles and the home hub to users in bulk.	Assignments saved; drivers require a home hub.	VERIFIED

ID	P	Requirement	Expected result	Label
FR-USR-005	P2	Users shall update their own name and phone and change their password.	Profile saved after validation (name 2–60 characters).	VERIFIED

Role and Permission Management (RBAC)

Detail: **BM-FRD-RBAC** (Functional Requirements — Role and Permission Management)

ID	P	Requirement	Expected result	Label
FR-RBAC-001	P1	The system shall provide 13 system roles with default permissions, data scopes and landing pages.	Roles listed with permission counts; locked roles cannot be deleted.	VERIFIED
FR-RBAC-002	P1	Administrators shall create custom roles with a name, data scope, landing page and any set of the 110 permission keys.	Users with the custom role see only permitted menus and a workspace launcher dashboard.	VERIFIED
FR-RBAC-003	P1	Administrators shall view and edit the permission matrix for editable roles.	Matrix changes take effect immediately in menus and services.	VERIFIED
FR-RBAC-004	P1	Every business action shall be checked against the user's permissions in the service layer, not only by hiding buttons.	Service returns a permission error when called without the key.	VERIFIED
FR-RBAC-005	P1	Lists shall be limited by data scope: own records (Salesman), linked customer (Customer), own deliveries (Driver), etc.	Salesman sees only own orders; driver cannot open another driver's delivery.	VERIFIED
FR-RBAC-006	P1	Production APIs shall enforce the same permission keys and data scopes server-side.	API returns 403 for missing permission and 404 for out-of-scope records.	PROPOSED

Dashboards and Analytics (DASH)

Detail: **BM-FRD-DASH** (Functional Requirements — Dashboards and Analytics)

ID	P	Requirement	Expected result	Label
FR-DASH-001	P1	Each role shall see a dashboard with the KPIs and work queues for that role.	Dashboard component matches the role (Admin, Salesman, Sales Manager, CS, Dispatcher, Finance, Warehouse, QA, Supplier).	VERIFIED
FR-DASH-002	P1	Every dashboard shall offer period presets (Today … All Time), custom range and specific periods including the April–March financial year.	KPIs, tables and charts follow the selected period.	VERIFIED
FR-DASH-003	P2	Dashboards shall optionally compare the selected period with the previous equivalent period.	KPI shows % change, difference and previous value.	VERIFIED
FR-DASH-004	P2	Current positions (stock, credit balance, fleet) shall be labelled Current and not filtered by period.	"Current" label shown on those KPIs.	VERIFIED
FR-DASH-005	P2	KPI cards and queue rows shall drill down into the filtered module list.	Clicking "Pending approval" opens the approval queue.	VERIFIED

ID	P	Requirement	Expected result	Label
FR-DASH-006	P2	Management shall have a live order monitor with real-time status updates (server push).	Status changes appear without page refresh.	PROPOSED

Customer Management (CUS)

Detail: **BM-FRD-CUS** (Functional Requirements — Customer Management)

ID	P	Requirement	Expected result	Label
FR-CUS-001	P1	Administrators shall create customers with code, type, contacts, address, tax IDs, hub, zone, sales representative, VIP flag and credit settings.	Customer saved and available for ordering.	VERIFIED
FR-CUS-002	P1	Customer data shall be validated (unique code, 10-digit mobile, email, PAN AAAAA9999A, GSTIN containing PAN, 6-digit pincode).	Invalid fields show messages; record is not saved.	VERIFIED
FR-CUS-003	P2	Users shall view customer details with credit position, orders, deliveries and documents within their scope.	Salesman sees only assigned customers' documents.	VERIFIED
FR-CUS-004	P2	Customers referenced by orders or other records shall not be deletable.	Delete is blocked and the blocking records are listed.	VERIFIED
FR-CUS-005	P1	Customer address and tax fields shall follow the operating country (UAE: emirate, TRN, +971 vs India: pincode, GSTIN/PAN, +91).	Client confirms country rules before production.	PENDING CONFIRMATION

Product Catalogue (PRD)

Detail: **BM-FRD-PRD** (Functional Requirements — Product Catalogue)

ID	P	Requirement	Expected result	Label
FR-PRD-001	P1	Administrators shall maintain products with code, SKU, barcode, category, unit, price, tax %, maximum discount, storage condition and stock settings.	Product saved and orderable when Active.	VERIFIED
FR-PRD-002	P1	Product codes, SKUs and barcodes shall be unique; tax 0–100%; maximum stock greater than minimum.	Duplicate or invalid values are rejected.	VERIFIED
FR-PRD-003	P2	Inactive or discontinued products shall not be orderable and shall block approval of orders containing them.	Approval check "Products active" fails.	VERIFIED
FR-PRD-004	P2	Item master and quantities shall be synchronised from SAP in stage two (today updated manually from emails).	Items updated automatically from SAP.	PROPOSED

Hubs, Suppliers and Organisation Master (HUB)

Detail: **BM-FRD-HUB** (Functional Requirements — Hubs, Suppliers and Organisation Master)

ID	P	Requirement	Expected result	Label
FR-HUB-001	P1	Administrators shall maintain hubs (type, address, contacts, status).	Hub available for stock, orders and dispatch when Active.	VERIFIED
FR-HUB-002	P2	Administrators shall maintain suppliers with lead time and products for replenishment.	Supplier selectable on replenishment requests.	VERIFIED
FR-HUB-003	P3	Administrators shall maintain the company profile and organisation structure (branches, business units, departments, designations, employees, warehouses).	Records saved and used on documents.	VERIFIED

Salesman Order Management (ORD)

Detail: **BM-FRD-ORD** (Functional Requirements — Salesman Order Management)

ID	P	Requirement	Expected result	Label
FR-ORD-001	P1	Salesmen shall create orders for a customer and hub with order type, priority, payment terms, order and requested delivery dates, delivery address and product lines.	Order saved as DRAFT with totals (subtotal, discount, tax, total).	VERIFIED
FR-ORD-002	P1	Salesmen shall submit orders for approval; the system assigns an order number and notifies the Sales Manager.	Status SUBMITTED, approval PENDING, number BM-ORD-..., notification “New order for approval”.	VERIFIED
FR-ORD-003	P1	The system shall validate order data: customer, salesman, hub, type, priority, payment terms, valid dates, address, 6-digit pincode, at least one line, total > 0.	Errors shown per field; order not submitted.	VERIFIED
FR-ORD-004	P1	The system shall show credit and stock warnings at submission without blocking it.	Warning flags stored on the order and shown to the approver.	VERIFIED
FR-ORD-005	P1	Salesmen shall correct and resubmit returned orders; the section matching the return reason is highlighted.	Status returns to SUBMITTED; timeline records the resubmission.	VERIFIED
FR-ORD-006	P2	Only the creator’s draft orders shall be deletable; submitted orders are kept for audit.	“Only draft orders can be deleted...”	VERIFIED
FR-ORD-007	P2	Users shall duplicate an existing order into a new draft.	New DRAFT with copied lines.	VERIFIED
FR-ORD-008	P1	Every order shall have a timeline of all events (created, submitted, decided, consolidated, dispatched, delivered).	Timeline shows each event with user and time.	VERIFIED
FR-ORD-009	P2	Salesmen shall see only their own orders; other roles see orders according to their scope.	Order list filtered by scope.	VERIFIED
FR-ORD-010	P2	Cancelling a submitted order on customer request.	Not available; Sales Manager may reject with reason “Customer cancelled”. PENDING CONFIRMATION of the cancellation rule.	NOT IMPLEMENTED
FR-ORD-011	P2	Orders shall record the source channel (call, email, WhatsApp, customer app) as described in the SOW.	Customer-app orders are tagged customer_app; office channel field to be confirmed.	PENDING CONFIRMATION

Sales Manager Approval (APR)

Detail: **BM-FRD-APR** (Functional Requirements — Sales Manager Approval)

ID	P	Requirement	Expected result	Label
FR-APR-001	P1	Sales Managers shall see an approval queue of submitted orders, oldest first.	Queue lists SUBMITTED / PENDING orders.	VERIFIED
FR-APR-002	P1	The system shall run six checks: customer active, hub active, credit within limit, stock available, total valid, products active.	Checks displayed with pass / warn / block.	VERIFIED
FR-APR-003	P1	Sales Managers shall approve orders; warnings must be acknowledged and remarks entered when required by settings.	Status APPROVED, consolidation PENDING, Customer Service and salesman notified.	VERIFIED
FR-APR-004	P1	Sales Managers shall return orders for correction with a reason from a list and remarks.	Status RETURNED; salesman notified with edit link.	VERIFIED
FR-APR-005	P1	Sales Managers shall reject orders with a reason from a list and remarks.	Status REJECTED (final); salesman notified.	VERIFIED
FR-APR-006	P1	Draft or already-decided orders shall not be decided again.	"…has already been approved. No change was made."	VERIFIED
FR-APR-007	P1	The Sales Manager shall be able to set the approved value and quantity (SOW slide 8).	Not implemented: the prototype approves the order as submitted; return for correction is used instead.	PENDING CONFIRMATION
FR-APR-008	P1	Finance shall be notified of approved orders, flagged when the credit limit is exceeded (SOW slides 8 and 16).	Not implemented: approval notifies Customer Service and the salesman only.	NOT IMPLEMENTED

Credit Validation (CRD)

Detail: **BM-FRD-CRD** (Functional Requirements — Credit Validation)

ID	P	Requirement	Expected result	Label
FR-CRD-001	P1	The system shall calculate the projected balance (current balance + order total) against the credit limit for every order.	Credit position shown on order and approval screens.	VERIFIED
FR-CRD-002	P1	Exceeding the credit limit shall be a warning by default and a blocking check when "Block when credit exceeded" is enabled.	Approve is disabled when blocking is on and the limit is exceeded.	VERIFIED
FR-CRD-003	P2	Users shall view credit limits, balances and utilisation for all customers.	Customer Credit page lists utilisation and over-limit customers.	VERIFIED
FR-CRD-004	P1	Customer balances shall be updated from invoices and payments.	Not implemented: balance is a manually maintained field; payments are not recorded.	NOT IMPLEMENTED
FR-CRD-005	P2	Credit balances shall be synchronised from the finance system (SAP) in production.	Balance reflects ERP receivables.	PROPOSED

Order Consolidation (Customer Service) (CON)

Detail: **BM-FRD-CON** (Functional Requirements — Order Consolidation (Customer Service))

ID	P	Requirement	Expected result	Label
FR-CON-001	P1	Customer Service shall see approved orders grouped per salesman, oldest approval first.	Queue by salesman displayed.	VERIFIED
FR-CON-002	P1	Customer Service shall confirm five verification checks and review hub stock before consolidating.	Status CONSOLIDATED; Dispatcher notified.	VERIFIED
FR-CON-003	P1	Customer Service shall release consolidated orders to dispatch.	Status READY_FOR_DISPATCH; Dispatcher and salesman notified.	VERIFIED
FR-CON-004	P2	Only approved orders shall be consolidated and only once.	“Only approved orders can be consolidated.”	VERIFIED

Dispatch Management (DSP)

Detail: **BM-FRD-DSP** (Functional Requirements — Dispatch Management)

ID	P	Requirement	Expected result	Label
FR-DSP-001	P1	Dispatchers shall create dispatches from ready orders for a hub, zone and date, saved as draft or planned.	Dispatch BM-DSP-... in DRAFT or PLANNED.	VERIFIED
FR-DSP-002	P1	Orders not ready, already on another active dispatch or without a complete address shall be refused.	Specific refusal message per order.	VERIFIED
FR-DSP-003	P1	Dispatchers shall set the stop sequence by drag and drop or move buttons; a suggested sequence (priority, pincode) is offered.	Sequence saved; driver notified of changes.	VERIFIED
FR-DSP-004	P1	Dispatchers shall confirm loading after a review tick; stock is deducted once and one delivery is created per order.	Status CONFIRMED; orders DISPATCHED; Stock Dispatch movements.	VERIFIED
FR-DSP-005	P1	Dispatchers shall cancel dispatches with a reason before any stop starts; confirmed dispatches reverse stock once.	Status CANCELLED; orders back to READY_FOR_DISPATCH.	VERIFIED
FR-DSP-006	P2	Dispatches shall be editable only while DRAFT, PLANNED or ASSIGNED.	“...is confirmed and can no longer be edited.”	VERIFIED
FR-DSP-007	P1	Dispatch status shall close automatically: COMPLETED when all stops are delivered, PARTIALLY_DELIVERED (Closed with exceptions) otherwise.	Status updates when the last stop closes.	VERIFIED
FR-DSP-008	P1	Dispatchers shall place orders into zones by drag and drop on a zone board (SOW slide 10).	Prototype: zone chosen in the dispatch form; drag and drop exists for stop sequence only.	PENDING CONFIRMATION
FR-DSP-009	P1	The system shall suggest the fastest route using distance and live traffic and warn when a VIP requested time may be missed.	Prototype suggests sequence by priority and pincode only.	PROPOSED

Zone Management (ZON)

Detail: **BM-FRD-ZON** (Functional Requirements — Zone Management)

ID	P	Requirement	Expected result	Label
FR-ZON-001	P1	Administrators shall maintain zones per hub with a list of pincodes.	Zone saved; code unique; at least one pincode.	VERIFIED
FR-ZON-002	P2	The system shall warn when a dispatch mixes orders from non-neighbouring zones.	Neighbouring-zone warning shown.	VERIFIED

Driver and Vehicle Assignment (DRA)

Detail: **BM-FRD-DRA** (Functional Requirements — Driver and Vehicle Assignment)

ID	P	Requirement	Expected result	Label
FR-DRA-001	P1	Dispatchers shall assign a driver and a vehicle to a planned dispatch.	Status ASSIGNED; driver notified.	VERIFIED
FR-DRA-002	P1	Drivers shall be refused when inactive, busy on another active dispatch, offline, from another hub, or with an expired or missing licence.	Specific refusal messages.	VERIFIED
FR-DRA-003	P1	Vehicles shall be refused when inactive, in maintenance, busy, from another hub, over capacity, or with expired required documents.	“Load of N exceeds … capacity” etc.	VERIFIED
FR-DRA-004	P1	Administrators shall maintain drivers (licence, hub, zone, vehicle, app login) and vehicles (number, type, capacity, hub, documents).	Records validated (Indian vehicle number format, unique licence).	VERIFIED

Driver Application (DRV)

Detail: **BM-FRD-DRV** (Functional Requirements — Driver Application)

ID	P	Requirement	Expected result	Label
FR-DRV-001	P1	Drivers shall see today’s and tomorrow’s stops published by the Dispatcher, the next stop and their performance.	Today plan listed; tomorrow plan visible with Start disabled.	VERIFIED
FR-DRV-002	P1	Drivers shall start a stop only when the dispatch is confirmed, the date has arrived, no other stop is active and it is next in sequence.	Blocker message shown otherwise; status OUT_FOR_DELIVERY when allowed.	VERIFIED
FR-DRV-003	P1	Drivers shall record arrival at the customer, which issues the delivery OTP.	arrivedAt recorded; OTP issued (simulated).	VERIFIED
FR-DRV-004	P1	Drivers shall complete a stop with delivered / damaged quantities, receiver name, optional signature and photo, and the customer’s OTP.	DELIVERED or PARTIALLY_DELIVERED; delivery note generated.	VERIFIED
FR-DRV-005	P1	Drivers shall report a failed delivery with a reason from the list and remarks.	FAILED; Dispatcher and customer notified.	VERIFIED
FR-DRV-006	P2	Drivers shall view the route of their stops on a map.	DEMO route map displayed (simulated).	VERIFIED
FR-DRV-007	P2	Drivers shall call or message the customer contact from the stop screen.	Phone / WhatsApp links open the device app.	VERIFIED

ID	P	Requirement	Expected result	Label
FR-DRV-008	P2	Drivers shall view and upload their own documents (licence, medical).	Upload goes to the review queue.	VERIFIED
FR-DRV-009	P1	VIP stops shall show a VIP badge; the driver follows the sequence published by the Dispatcher.	VIP badge displayed.	VERIFIED
FR-DRV-010	P1	The production driver app shall be a React Native app with background GPS (10–15 s updates) during active deliveries only.	Location continues when the phone is locked; last update time shown.	PROPOSED
FR-DRV-011	P2	The production driver app shall queue status updates when offline and sync when connectivity returns.	No data loss in poor coverage.	PROPOSED

Customer Application (CAP)

Detail: **BM-FRD-CAP** (Functional Requirements — Customer Application)

ID	P	Requirement	Expected result	Label
FR-CAP-001	P1	Customers shall browse and search active products with prices and add them to a cart.	Cart persists per customer.	VERIFIED
FR-CAP-002	P1	Customers shall place orders with delivery address, requested date (tomorrow to 60 days), time slot, reference and remarks.	Order SUBMITTED with source customer_app; cart cleared.	VERIFIED
FR-CAP-003	P1	Customers shall follow each order with a stage tracker and customer-friendly labels.	Stages from Order placed to Delivered.	VERIFIED
FR-CAP-004	P1	Customers shall track their own delivery on a map and see the OTP when the driver arrives.	Only own order's vehicle is visible.	VERIFIED
FR-CAP-005	P2	Customers shall view and print delivery notes and invoices.	Documents listed for the linked customer only.	VERIFIED
FR-CAP-006	P2	Customers shall reorder a past order into the cart.	Orderable lines copied to cart.	VERIFIED
FR-CAP-007	P2	Customers shall pay online and see payment history.	Page shows balance and "No payments recorded in this prototype".	NOT IMPLEMENTED
FR-CAP-008	P2	The customer app is an addition beyond the SOW (which lists Order Taking App, Drivers App and Admin Panel); inclusion in the production scope needs confirmation.	Client decision recorded.	PENDING CONFIRMATION

Delivery Tracking (TRK)

Detail: **BM-FRD-TRK** (Functional Requirements — Delivery Tracking)

ID	P	Requirement	Expected result	Label
FR-TRK-001	P1	Office users shall track every stop on active dispatches with status and times.	List shows live statuses.	VERIFIED
FR-TRK-002	P2	Administrators and Dispatchers shall see vehicle positions on a fleet map.	DEMO GPS positions move along a straight-line route.	PROTOTYPE
FR-TRK-003	P1	Production tracking shall push real GPS positions to the admin map (all vehicles) and the customer map (own order only) over WSS.	Updates appear without refresh; customers never see the fleet.	PROPOSED

Delivery OTP (OTP)

Detail: **BM-FRD-OTP** (Functional Requirements — Delivery OTP)

ID	P	Requirement	Expected result	Label
FR-OTP-001	P1	A 6-digit OTP shall be issued on arrival, valid 10 minutes by default (1–60), shown to the customer.	OTP visible in the customer app; no SMS in the prototype.	PROTOTYPE
FR-OTP-002	P1	Completion shall require the correct, unexpired OTP; 5 wrong attempts lock the OTP.	Wrong OTP blocks completion; LOCKED after 5 attempts.	VERIFIED
FR-OTP-003	P2	Drivers shall generate a new OTP when it has expired or is locked; the previous one is invalidated.	New OTP ACTIVE; old INVALIDATED.	VERIFIED
FR-OTP-004	P2	Dispatchers shall be able to inspect a delivery's OTP status (dispatch.otp.inspect).	OTP state visible to the Dispatcher.	VERIFIED
FR-OTP-005	P1	Production OTPs shall be sent to the customer by SMS or WhatsApp and stored hashed.	Customer receives the code on their phone.	PROPOSED

Proof of Delivery (POD)

Detail: **BM-FRD-POD** (Functional Requirements — Proof of Delivery)

ID	P	Requirement	Expected result	Label
FR-POD-001	P1	Proof of delivery shall capture receiver name, optional signature and photo (≤ 350 KB each) and customer remarks (≤ 300 characters).	POD stored with the delivery.	VERIFIED
FR-POD-002	P1	Delivered quantity shall not exceed the dispatched quantity; damaged and rejected quantities are recorded.	Validation messages per line.	VERIFIED
FR-POD-003	P1	Production POD images shall be stored in object storage with signed URLs and retention rules.	No images in browser storage.	PROPOSED

Delivery Notes (DN)

Detail: **BM-FRD-DN** (Functional Requirements — Delivery Notes)

ID	P	Requirement	Expected result	Label
FR-DN-001	P1	A numbered delivery note (BM-DN-YYYY-NNNNN) shall be generated when a stop is completed (full or partial).	Note lists delivered, short and damaged quantities and receiver.	VERIFIED
FR-DN-002	P1	Delivery notes shall be printable and visible to office roles with permission, the driver and the customer.	Print layout available.	VERIFIED

ID	P	Requirement	Expected result	Label
FR-DN-003	P1	The delivery note shall be sent to the customer instantly by SMS / WhatsApp / email (SOW slide 13).	Prototype: visible in the customer app only.	PROPOSED

Delivery Exception Review (EXC)

Detail: **BM-FRD-EXC** (Functional Requirements — Delivery Exception Review)

ID	P	Requirement	Expected result	Label
FR-EXC-001	P1	Dispatchers shall review failed and partial deliveries with remarks.	Review status REVIEWED.	VERIFIED
FR-EXC-002	P1	The review may return undelivered (not damaged) stock to the hub once.	Delivery Return movement posted once.	VERIFIED
FR-EXC-003	P1	A failed delivery whose goods are returned may be re-queued for a new dispatch.	Order READY_FOR_DISPATCH.	VERIFIED

Inventory Management (INV)

Detail: **BM-FRD-INV** (Functional Requirements — Inventory Management)

ID	P	Requirement	Expected result	Label
FR-INV-001	P1	Warehouse users shall record goods receipts that increase hub stock immediately.	GRN created; Stock Receipt movement.	VERIFIED
FR-INV-002	P1	Warehouse users shall transfer stock between hubs as a draft or completed immediately.	Transfer Out / In movements on completion.	VERIFIED
FR-INV-003	P1	Authorised users shall adjust stock (increase, decrease, damaged, lost, correction) with a reason.	Movement posted immediately.	VERIFIED
FR-INV-004	P1	Every stock change shall be written to the movement ledger.	Ledger shows type, quantity, reference, user and time.	VERIFIED
FR-INV-005	P2	Users shall set minimum, reorder and maximum levels per stock line.	Low Stock list uses the reorder level.	VERIFIED
FR-INV-006	P1	Stock cannot go negative unless the product allows it; completed transfers are not posted twice.	“Only N available…” / “already completed”.	VERIFIED
FR-INV-007	P2	Batch and expiry tracking per stock line, FEFO picking and cycle counting.	Settings exist (UI only); not enforced.	NOT IMPLEMENTED
FR-INV-008	P2	Stock positions shall be synchronised with SAP in stage two.	Stock reconciled with ERP.	PROPOSED

Scrap Management (SCR)

Detail: **BM-FRD-SCR** (Functional Requirements — Scrap Management)

ID	P	Requirement	Expected result	Label
FR-SCR-001	P1	Users shall raise scrap requests with reason, quantity, stock source, batch and photo evidence.	DRAFT or PENDING_APPROVAL.	VERIFIED

ID	P	Requirement	Expected result	Label
FR-SCR-002	P1	Quantity shall not exceed eligible stock; photo required for Damaged, Packaging Damage and Contamination; batch required for batch-controlled products; no duplicate open request.	Validation messages.	VERIFIED
FR-SCR-003	P1	Approvers shall approve or reject requests; requesters cannot approve their own.	APPROVED / REJECTED.	VERIFIED
FR-SCR-004	P1	Approvers shall post the stock write-off exactly once.	COMPLETED; Scrap Write-off movement.	VERIFIED
FR-SCR-005	P3	A printable scrap note shall be available.	Print layout.	VERIFIED

Warehouse and QA Dashboards (WQA)

Detail: **BM-FRD-WQA** (Functional Requirements — Warehouse and QA Dashboards)

ID	P	Requirement	Expected result	Label
FR-WQA-001	P2	Warehouse users shall see picking & loading, inventory alerts, damaged stock and document expiry.	Dashboard lists consolidated / ready orders per hub.	VERIFIED
FR-WQA-002	P2	QA users shall see inspection queues, quarantine, expired documents and the cold-chain card.	Cold-chain card states “No sensor integration”.	VERIFIED
FR-WQA-003	P2	Live cold-chain temperature monitoring and excursion alerts.	Not available (ranges configured only).	NOT IMPLEMENTED

Vehicle Management System (VMS)

Detail: **BM-FRD-VMS** (Functional Requirements — Vehicle Management System)

ID	P	Requirement	Expected result	Label
FR-VMS-001	P1	The system shall compute compliance per vehicle and driver document (Valid, 30 days, 7 days, today, expired, missing).	Expiry register sorted by urgency.	VERIFIED
FR-VMS-002	P1	Administrators shall define required documents and blocking rules; an expired required document always blocks assignment.	Assignment blocked with the specific document message.	VERIFIED
FR-VMS-003	P2	Authorised users shall upload vehicle and driver documents with expiry dates.	Document enters the review flow.	VERIFIED
FR-VMS-004	P2	Opening the expiry register shall raise DOCUMENT_EXPIRING / DOCUMENT_EXPIRED events.	Automated notification when a rule is enabled.	VERIFIED
FR-VMS-005	P3	Fuel, maintenance and trip cost tracking.	Related reports show “no source data”.	NOT IMPLEMENTED

Document Management and OCR (DOC)

Detail: **BM-FRD-DOC** (Functional Requirements — Document Management and OCR)

ID	P	Requirement	Expected result	Label
FR-DOC-001	P1	Users shall upload documents for records they may access, with type, linked record, file and key fields.	Status PENDING_OCR, PENDING_ REVIEW or VERIFIED by type settings.	VERIFIED
FR-DOC-002	P1	The system shall extract fields with OCR and let reviewers correct them (originals kept).	MOCK OCR pre-fills from the linked record (labelled Simulated OCR).	PROTOTYPE
FR-DOC-003	P1	Reviewers shall verify or reject documents; rejection requires a reason.	VERIFIED (locked) or REJECTED.	VERIFIED
FR-DOC-004	P2	Administrators shall maintain document types (fields, expiry, OCR, verification).	Built-in VMS codes cannot be changed.	VERIFIED
FR-DOC-005	P1	Production OCR shall run server-side (e.g. Tesseract / cloud OCR) with virus scanning of uploads.	Real field extraction; files never executed.	PROPOSED

Invoice Customisation and Invoicing (INVC)

Detail: **BM-FRD-INVC** (Functional Requirements — Invoice Customisation and Invoicing)

ID	P	Requirement	Expected result	Label
FR-INVC-001	P2	Finance shall maintain invoice templates (layout, fonts, colours, sections, default).	Template preview available.	VERIFIED
FR-INVC-002	P2	Finance shall issue one invoice per approved-or-later order on ordered or delivered quantities with discounts and charges.	Invoice BM-INV-YYYY-NNNNN ISSUED with frozen snapshot.	VERIFIED
FR-INVC-003	P2	Finance shall cancel an invoice with a reason, after which a new invoice can be issued.	CANCELLED; order invoiceable again.	VERIFIED
FR-INVC-004	P2	Finance shall configure invoice numbering (prefix, year, digits).	Next number follows settings.	VERIFIED
FR-INVC-005	P1	The SOW states invoicing stays outside the application; the prototype includes invoicing. Production scope must be confirmed.	Client decision recorded.	PENDING CONFIRMATION
FR-INVC-006	P2	Payments, receipts, credit notes and refunds.	Not available.	NOT IMPLEMENTED

Email Communication (EML)

Detail: **BM-FRD-EML** (Functional Requirements — Email Communication)

ID	P	Requirement	Expected result	Label
FR-EML-001	P2	Users shall compose and send emails with templates and document attachments.	Demo mode records SENT_DEMO (“not sent”).	PROTOTYPE
FR-EML-002	P2	Administrators shall maintain email templates with variables.	Templates render with record values.	VERIFIED
FR-EML-003	P3	Users shall read the mailbox.	Seeded demo mailbox.	PROTOTYPE
FR-EML-004	P2	Live SMTP / IMAP through the integration gateway.	Not verified (no gateway available).	NOT VERIFIED

OpenWA (WhatsApp) Communication (OWA)

Detail: **BM-FRD-OWA** (Functional Requirements — OpenWA (WhatsApp) Communication)

ID	P	Requirement	Expected result	Label
FR-OWA-001	P2	Users shall send WhatsApp messages using templates.	Mock provider records "Sent · demo".	PROTOTYPE
FR-OWA-002	P2	Live OpenWA session (QR, start / stop) through the gateway.	Not verified.	NOT VERIFIED
FR-OWA-003	P1	The customer messaging provider (SMS or WhatsApp; OpenWA is unofficial) must be confirmed at kickoff (SOW slide 16).	Decision recorded in the decision log.	PENDING CONFIRMATION

Automated Notifications (AUTO)

Detail: **BM-FRD-AUTO** (Functional Requirements — Automated Notifications)

ID	P	Requirement	Expected result	Label
FR-AUTO-001	P2	Administrators shall configure per-event rules (channel, recipients, templates) and a master switch.	11 events configurable.	VERIFIED
FR-AUTO-002	P2	Workflow events shall trigger messages once per record (de-duplicated) and be logged in history.	History shows SENT_DEMO / SKIPPED with reason.	VERIFIED

Call History (CALL)

Detail: **BM-FRD-CALL** (Functional Requirements — Call History)

ID	P	Requirement	Expected result	Label
FR-CALL-001	P3	Users shall log calls with direction, status, outcome, duration and notes.	Call saved; salesmen see their own calls only.	VERIFIED
FR-CALL-002	P3	Users shall schedule and complete follow-ups and link orders to calls.	Follow-ups grouped overdue / today / upcoming.	VERIFIED

Notifications (in-app) (NOTF)

Detail: **BM-FRD-NOTF** (Functional Requirements — Notifications (in-app))

ID	P	Requirement	Expected result	Label
FR-NOTF-001	P1	Users shall receive in-app notifications for workflow events addressed to their role, user, driver or customer record.	Bell count and Notification Center.	VERIFIED
FR-NOTF-002	P3	Users shall mark notifications read and mute categories.	Preferences saved per user.	VERIFIED
FR-NOTF-003	P2	Production apps shall send push notifications to drivers and customers.	Push received on the phone.	PROPOSED

Reports (RPT)

Detail: **BM-FRD-RPT** (Functional Requirements — Reports)

ID	P	Requirement	Expected result	Label
FR-RPT-001	P2	Users shall browse 118 reports in 11 categories filtered by permission, with search, favourites and recent.	Only permitted categories visible.	VERIFIED
FR-RPT-002	P2	Reports shall run with date, month, year and grouping filters within the user's data scope and export to CSV or print.	CSV downloaded; print layout.	VERIFIED
FR-RPT-003	P3	Reports without source data shall say so with the reason.	24 reports show "no source data".	VERIFIED

AI Assistant, Forecasting and Replenishment (AI)

Detail: **BM-FRD-AI** (Functional Requirements — AI Assistant, Forecasting and Replenishment)

ID	P	Requirement	Expected result	Label
FR-AI-001	P3	Users shall ask business questions in natural language and receive tables and charts from permission-checked tools.	Demo intent engine answers; write actions need confirmation.	PROTOTYPE
FR-AI-002	P3	Users shall generate demand forecasts with selectable models and accuracy metrics.	Forecast saved and exportable.	VERIFIED
FR-AI-003	P2	Users shall raise replenishment requests; Administrators approve, reject and convert them; stock receipts complete them.	DRAFT → SUBMITTED → APPROVED → CONVERTED → RECEIVED.	VERIFIED
FR-AI-004	P3	The system shall generate rule-based insights that users mark reviewed or resolved.	NEW → REVIEWED → RESOLVED.	VERIFIED
FR-AI-005	P3	External AI providers (OpenJarvis, Ollama, Gemini, OpenAI-compatible).	Not verified.	NOT VERIFIED

System Settings and Audit (SET)

Detail: **BM-FRD-SET** (Functional Requirements — System Settings and Audit)

ID	P	Requirement	Expected result	Label
FR-SET-001	P1	Administrators shall configure security, approval, localisation, numbering, inventory, warehouse and cold-chain settings.	Settings validated and applied.	VERIFIED
FR-SET-002	P1	Administrative and security events shall be recorded in the audit log.	User, role, settings and lockout events listed.	VERIFIED
FR-SET-003	P2	Administrators shall reset demo data and export / import the application data.	Data reset to seed; JSON export validated on import.	VERIFIED
FR-SET-004	P1	Production data shall be stored in PostgreSQL with daily backups, point-in-time recovery and audit fields on every table.	Data shared across users and devices.	PROPOSED

Revision history

Version	Date	Author	Description
1.0	2026-11-12	Documentation team	Initial draft prepared from analysis of the bellomassi-premium-ui source code and existing project documents.